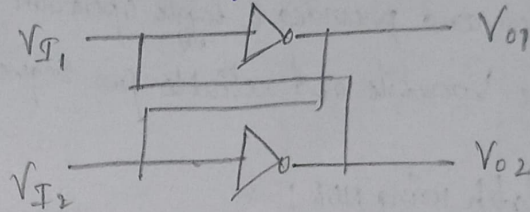
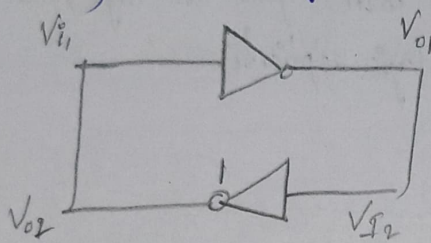


3) Define behaviour of bistable operation of sequential logic using Voltage transfer char!

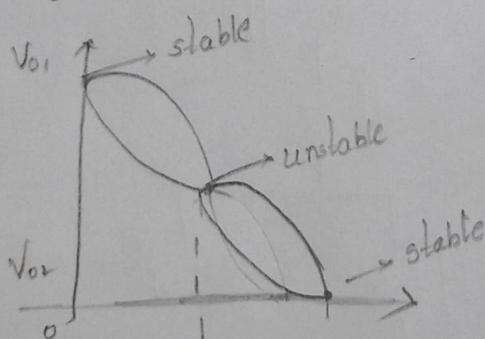
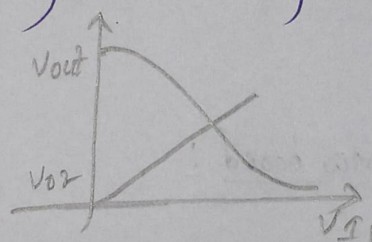
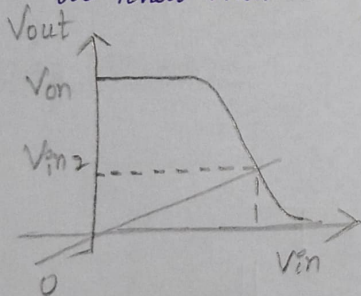
Ans:

Bistable operation is the property of a sequential logic circuit to have two stable equilibrium states which can be explained using voltage transfer characteristic where the intersection of VTC curves results in two stable points and one unstable point, enabling the circuit to store binary information (0 or 1).

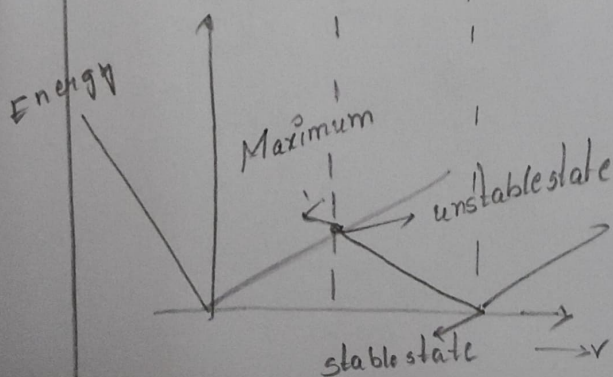


From the above bistable element we can move $V_{i1} = V_{o2}$, $V_{i2} = V_{o1}$ w.k.t Voltage transfer characteristic from inverter.

We know that Voltage transfer characteristics for Inverter.



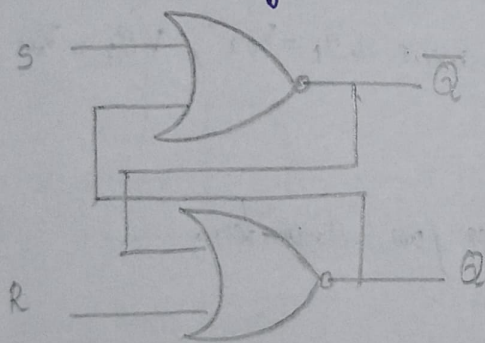
There are 2 stable state w.r.t logic (0 & 1)
electrical (Gnd - VDD) with logics of
(logic 0 & logic 1)



2) Analyse the operation of MOS based SR latch Circuit based on 2T1p NOR gates and clocked JK flipflop over clocked RS flipflop?

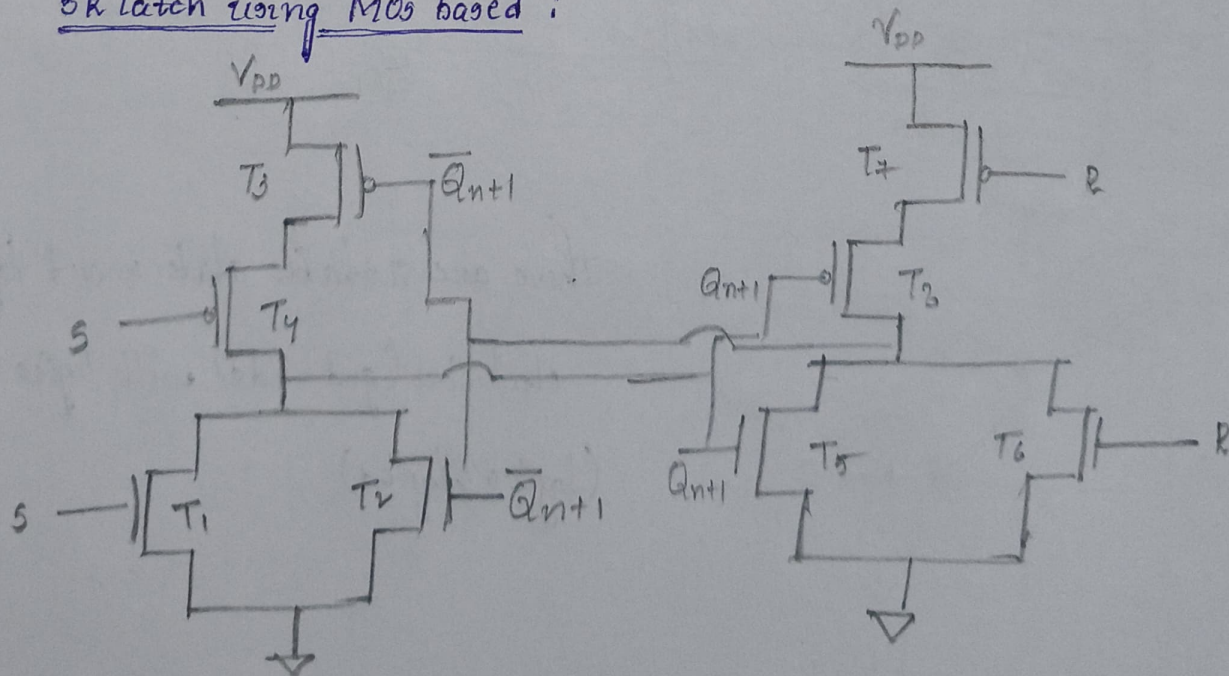
A MOS-based SR latch using two NOR gate operates as a bistable circuit storing one bit using cross-coupled feedback. However, it suffers from an invalid input condition ($S=R=1$). The clocked RS flipflop improves control by introducing a clocking signal but still retains the invalid state. The clocked JK flipflop overcomes this limitation by using feedback to eliminate the invalid condition and provides a toggle operation when both inputs are high, making it more versatile and reliable for sequential logic application.

1) SR latch using NOR : $\rightarrow$



S	R	Q	$\bar{Q}$
0	0	No change	
0	1	0	1
1	0	1	0
1	1	Invalid	

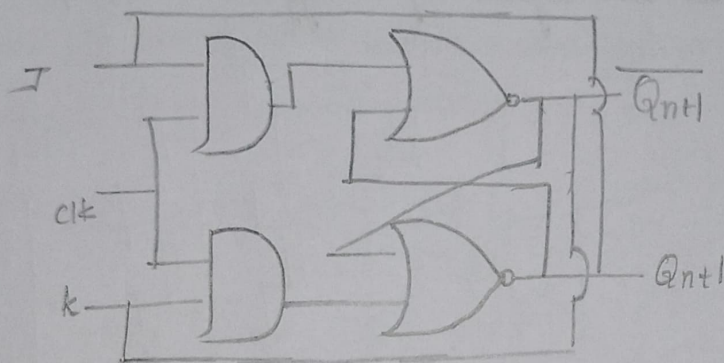
SR latch using MOS based :



S	R	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇	T ₈	Q _{n+1}	$\overline{Q}_{n+1}$
1	0	on	on	off	off	off	off	on	on	1	0
0	1	off	off	on	on	on	on	off	off	0	1
0	0	off	off	on	on	on	on	off	off	0	1
1	1	→ Invalid →									

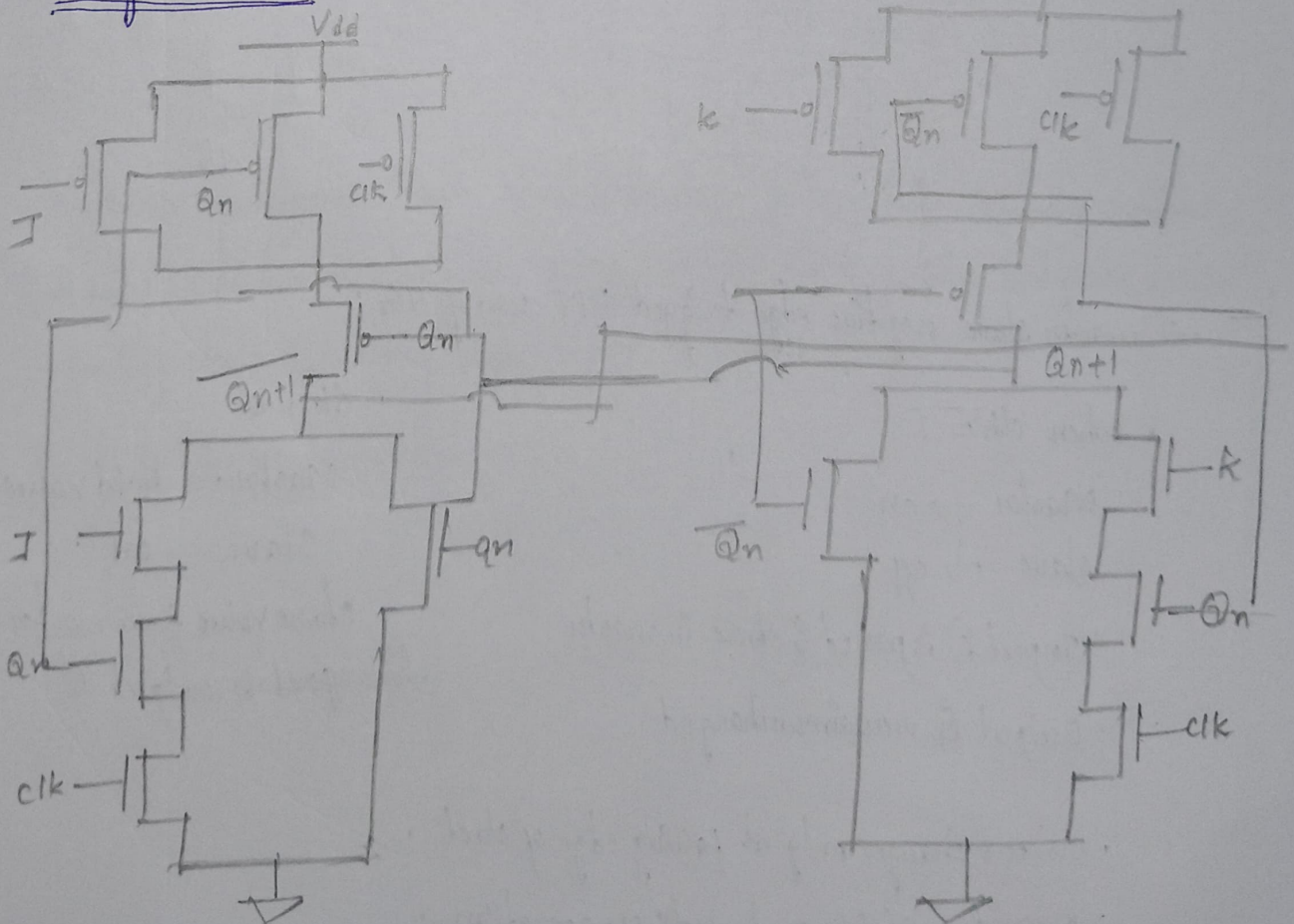
Note: To overcome that invalid case in the SR latch we prefer the JK latch, JK flipflop.

JK flipflop clocked using 2 NOR gates



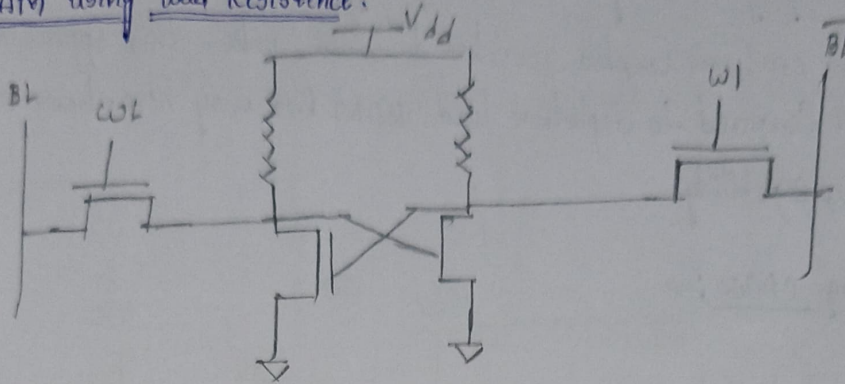
clk	J	K	Q _{n+1}	$\overline{Q}_{n+1}$
0	0	0	Q _n	Q _n
1	0	0	Q _n	Q _n
1	0	1	0	1
1	1	0	1	1
1	1	1	$\overline{Q}_n$	Q _n

using Mos based



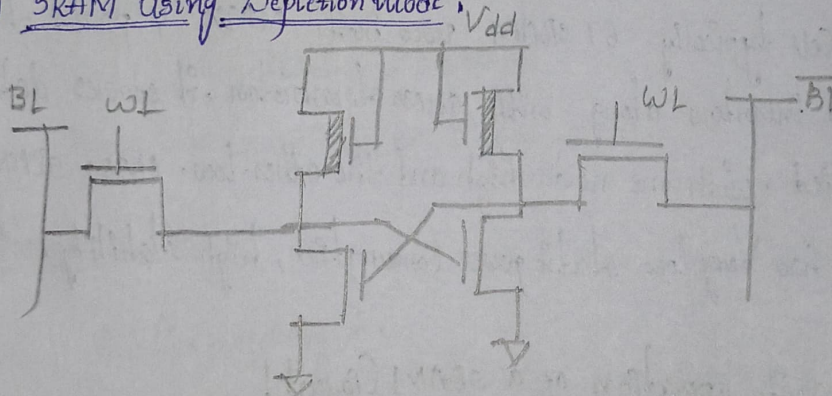
4. Discuss the working of Alternate form of SRAM Cell?

1) SRAM using Load Resistance:



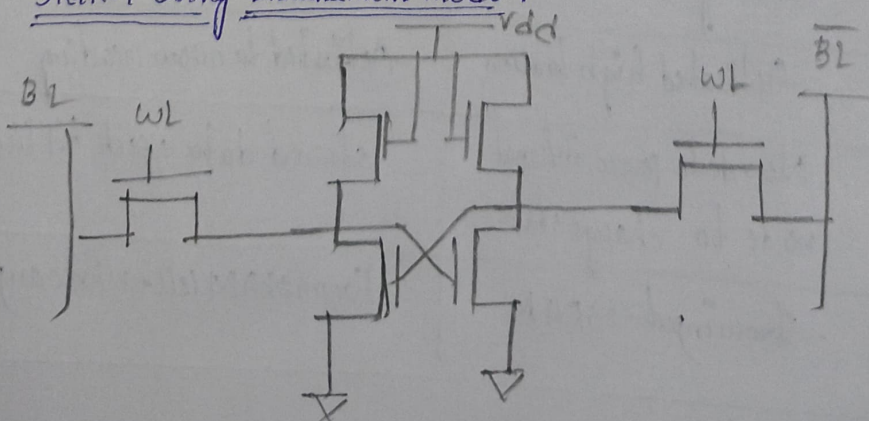
This is the type of SRAM cell, resistors are used as load element in place of active transistors. The cell consists of cross-coupled inverters where pull-up is provided by resistors and pull-down by NMOS transistors where value is stored, one transistor conducts while the other is off, maintaining either logic '0' or '1'. Although the design is simple.

2) SRAM using Depletion mode:



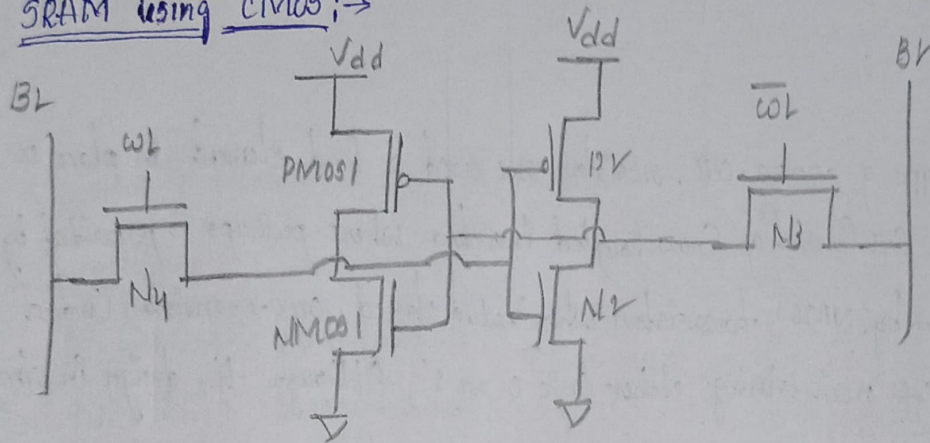
The SRAM uses depletion mode NMOS transistors as load devices instead of resistors. These transistors are always on, acting like high-value resistors but more efficient. The cell stores data using cross-coupled inverters.

3) SRAM Using Enhancement mode:



This is enhancement mode NMOS transistor are used as load device, these loads are Controlled by gate Voltage and not always ON like depletion loads. the operation still relies on Cross Coupled inverters to store data, this approach reduces power Consumption Compared to depletion load SRAM but may have slower switching and weaker pull-up Capability.

4) SRAM using CMOS :->



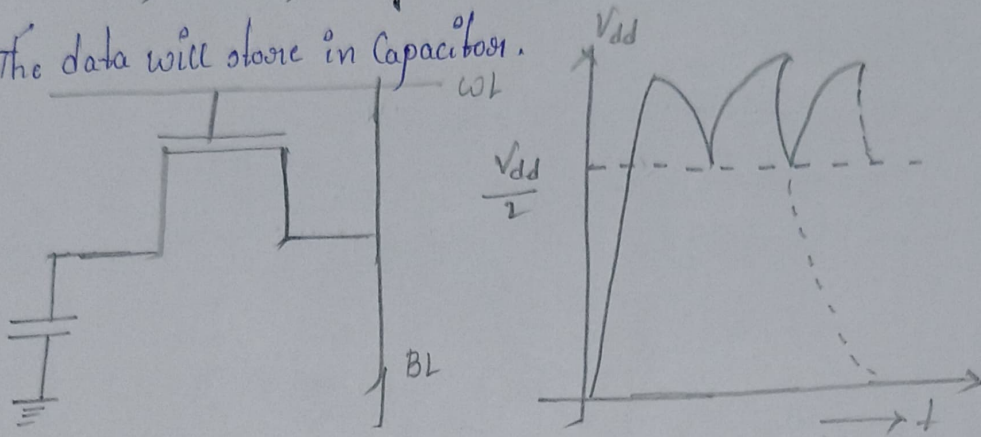
The CMOS SRAM Cell typically 6T1SRAM uses both NMOS & PMOS transistor to form Cross-Coupled inverters along with access transistor. it stores data as long as power is supplied, with one node high and the other low. CMOS SRAM is highly efficient because it has Very low static power Consumption, high stability.

5) Analyze read and write operation of a SRAM Circuit!

Features	Write (opr ⁿ)	Read (opr ⁿ)
purpose	store data (0 or 1)	Retrieve stored data
Bit line (BL, $\overline{BL}$)	Driven by external data	precharged
word line	Activated high to allow	Activated to allow reading
Action in cell	New data force internal node to change state	stored data effect bit line
Data flow	From input $\rightarrow$ SRAM	From SRAM cell $\rightarrow$ Sense amplifier

6) Explain the operation of a dynamic RAM Circuit?

Ans: The data will store in Capacitor.



- To reduce transistor we came to Dram

Write operation:

- Bit line Carries data (0 or 1).
- $W_L = \text{High}$ transistor ON
- Capacitor, charge stores $\rightarrow 1$
Discharge stores $\rightarrow 0$
- $W_L = \text{low}$ transistor off Data stored.

Read operation:

- Bit line precharged.
- $W_L = \text{High}$ Capacitor shares charge with bitline.
- Small voltage change occur on bitline.
- Sense amplifier detects whether it is 0 or 1.